

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Belagavi-560018, Karnataka, India



Internship Report On

“ANDROID APP DEVELOPMENT USING GEN-AI”

*Submitted in partial fulfilment of the requirement for the award of Degree of
Bachelor of Engineering in Computer Science and Engineering*

Submitted by

ADITYA ANJAN JHA

1VE22CS006

Carried out at

MINDMATRIX

HSR Layout, Bengaluru

Under the Guidance of

Internal Guide

Dr. Bama S

Associate Professor

Department of CSE

SVCE, Bengaluru

External Guide

Mr. Tirumala

Mutalikdesai,

Program Manager

MINDMATRIX, Bengaluru



SRI VENKATESHWARA

COLLEGE OF ENGINEERING

ESTD. 2001. AUTONOMOUS INSTITUTE

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

SRI VENKATESHWARA COLLEGE OF ENGINEERING

Vidyanagar, Bengaluru-562157

2025-2026

SRI VENKATESHWARA COLLEGE OF ENGINEERING

(Affiliated by A.I.C.T.E and approved by V.T.U)



SRI VENKATESHWARA
COLLEGE OF ENGINEERING
ESTD. 2001. AUTONOMOUS INSTITUTE

CERTIFICATE

This is to certify that the Internship work entitled **“ANDROID APP DEVELOPMENT USING GEN-AI”** has been successfully carried out by **ADITYA ANJAN JHA** bearing USN **1VE22CS006** a Bonafide student of the VIII semester in the Department of Computer science and Engineering, **Sri Venkateshwara College of Engineering, Bengaluru**, in partial fulfillment of the requirements for the award of degree of Bachelor of Engineering in Computer Science and Engineering as prescribed by **Visvesvaraya Technological University, Belgaum**, during the academic year **2025-26**. It is certified that all corrections and suggestions indicated for internship internal assessment have been duly incorporated into the report. The Internship report has been approved as it satisfies the academic requirements in respect of internship work prescribed for the said degree.

.....
Signature of Guide
Dr. BAMA S
Assoc. Prof,
Dept. of CSE SVCE,
Bengaluru

.....
Signature of Coordinator
Mr. ABHILASH D C
Asst. Prof,
Dept. of CSE SVCE,
Bengaluru

.....
Signature of HOD
Dr. MUKESH KUMAR SINGH
Dept. of CSE
SVCE, Bengaluru

.....
Signature of Principal
Dr. B M SATISH
SVCE,
Bengaluru

Name of the Examiners

- 1.
- 2.

Signature with Date

CERTIFICATE OF COMPLETION

SRI VENKATESHWARA COLLEGE OF ENGINEERING

(Affiliated by A.I.C.T.E and approved by V.T.U)



SRI VENKATESHWARA
COLLEGE OF ENGINEERING
ESTD. 2001. AUTONOMOUS INSTITUTE

DECLARATION

I, **ADITYA ANJAN JHA** bearing USN **1VE22CS006**, student of Bachelor of Engineering, **Department of Computer Science and Engineering, Sri Venkateshwara College of Engineering, Bengaluru**, hereby declare that the internship report entitled “**ANDROID APP DEVELOPMENT USING GEN-AI**” has been carried out by me under the supervision and guidance of **Dr. BAMA S**, Assoc. Prof, Department of CSE; **Mr. ABHILASH D C**, Internship Coordinator and Asst. Prof, Department of CSE; **Dr. MUKESH KUMAR SINGH**, HOD, Department of CSE; and **Mr. TIRUMALA MUTALIKDESAI**, Program Manager, **Mindmatrix** This report is submitted by me as a partial fulfillment for the award of **Bachelor of Engineering** degree in Computer Science and Engineering from **Visvesvaraya Technological University, Belagavi** during the academic year **2025-26**.

I hereby declare that the internship work carried out and submitted is entirely my own, and it has not been submitted previously for the award of any degree at any other institute or university.

Date: 10/05/2026

Place: Bengaluru

ADITYA ANJAN JHA

1VE22CS006

ABSTRACT

The Namma-Skill application is an AI-powered Android platform developed to support skill development, training accessibility, and self-employment opportunities for users. Traditional methods of accessing skill training programs often lack personalization, easy navigation, and centralized access to course information. This project addresses these challenges by providing a modern mobile application that enables users to explore courses, search training centers, receive recommendations, and manage enrollments efficiently through an intuitive digital platform.

The application was developed using Kotlin and Jetpack Compose in Android Studio following the MVVM architectural pattern for structured and scalable development. Firebase services were integrated for authentication, database management, and notifications, while Generative AI features were incorporated to provide intelligent course recommendations and smart user assistance. The system includes functionalities such as user registration and login, course browsing, category-based filtering, search functionality, notifications, profile management, and location-based training center support.

The frontend was designed with a responsive and user-friendly interface to ensure smooth navigation and improved user experience. The modular architecture of the application allows efficient integration between the Android application layer, database layer, AI module, and external services such as Firebase Cloud Messaging and Google Maps API.

The project was successfully tested within the Android environment, and the results demonstrated stable application performance, effective navigation, reliable data handling, and seamless user interaction. Overall, the Namma-Skill application provides a scalable and practical solution for digital learning and self-employment support while showcasing the integration of modern Android development technologies with artificial intelligence features.

ADITYA ANJAN JHA

1VE22CS006

ACKNOWLEDGEMENT

The satisfaction and euphoria that accompany the successful completion of any task would be incomplete without complementing those who made it possible, whose guidance and encouragement made my efforts successful.

I would like to express my sincere gratitude to the highly esteemed institution **SRI VENKATESHWARA COLLEGE OF ENGINEERING**, for providing me with the knowledge and resources necessary to pursue my journey toward becoming a software engineer.

I extend my heartfelt thanks to **Dr. B M SATISH**, Principal, SVCE, Bengaluru, for providing the required facilities.

I am extremely thankful to **Dr. MUKESH KUMAR SINGH**, HOD, Department of CSE, SVCE, for providing support and encouragement.

I would like to extend my heartfelt thanks to **Mr. ABHILASH D C**, Assistant Professor & Internship Coordinator, Department of CSE, SVCE, for his continuous support and coordination during my internship.

I am especially grateful to **Dr. BAMA S**, Associate Professor, Department of CSE, SVCE, for her invaluable guidance, encouragement, and constructive suggestions throughout the internship. Her support was essential for the successful completion of this work. I also thank all the computer science staff and others who helped me directly or indirectly to meet my internship with grand success.

I am also sincerely grateful to **Mr. TIRUMALA MUTALIKDESAI**, my external guide at **Mindmatrix** for his invaluable guidance, insightful feedback, and support during the course of the internship, which contributed greatly to the successful completion of this work.

Finally, I am grateful to my parents and friends for their support, guidance and encouragement.

ADITYA ANJAN JHA

1VE22CS006

INSTITUTE VISION

To be globally recognized institution of engineering excellence, innovation, and societal impact.

INSTITUTE MISSION

M1: Deliver transformative, outcome-based engineering education aligned with global standards through innovative curricula, state-of-the-art infrastructure, and learner-centric teaching.

M2: Advance interdisciplinary research and technological innovation that address real-world challenges, achieving national prominence and global visibility in engineering education.

M3: Strengthen collaborations with industry, government, and leading international universities to enhance professional competencies, entrepreneurship, and global employability.

M4: Nurture ethical, socially responsible, and environmentally conscious professionals committed to building sustainable solutions for society.

M5: Foster adaptability, curiosity, and lifelong learning to prepare graduates for emerging technologies and the dynamic global landscape.

DEPARTMENT VISION

To evolve as a Recognized Course in computer science and engineering education, nurturing competent professionals, innovators, and researchers who contribute ethically to technological advancement and sustainable societal development.

DEPARTMENT MISSION

M1: To impart high-quality, outcome-based computer science and engineering education through innovative curricula, advanced laboratories, and learner-focused pedagogies that meet industry standards.

M2: To promote cutting-edge research, interdisciplinary projects, and entrepreneurial initiatives that addresses real-world problems and contributes to technological growth and national development.

M3: To establish strong partnerships with industry, research organizations, and international institutions to enhance experiential learning, professional employability.

M4: To inculcate ethical values, social responsibility, and environmental awareness in graduates, enabling them to design sustainable computing solutions for the betterment of society.

M5: Nurture the essential skills of critical thinking, problem-solving, and continuous learning to equip graduates with the adaptability required to master emerging technologies and excel in the rapidly revolving landscape of information technology and digital transformation.

PROGRAM EDUCATIONAL OBJECTIVES

Knowledge:

Computer Science and Engineering Graduates will have professional technical career in interdisciplinary domains providing innovative and sustainable solutions using modern tools.

Skills:

Computer Science and Engineering Graduates will have effective communication, leadership, team building, problem solving, decision making and creative skills.

Attitude:

Computer Science and Engineering Graduates will practice ethical responsibilities towards their peers, employers and society.

PROGRAM SPECIFIC OUTCOMES

PSO 1:

Ability to adopt quickly for any domain, interact with diverse group of individuals and be an entrepreneur in a societal and global setting.

PSO 2:

Ability to visualize the operations of existing and future software Applications

TABLE OF CONTENTS

CHAPTER NO.	PAGE NO.
1. COMPANY PROFILE	01-03
1.1 Introduction	01
1.2 Mission	01
1.3 Vision	02
1.4 Services	02
1.5 Conclusion	03
2. ABOUT COMPANY DEPARTMENT	04
2.1 Overview of the Department	04
2.2 Organizational Structure	04
3. TASKS PERFORMED	06-18
3.1 Work Schedule	06
3.2 Introduction	07
3.3 System Requirements	08
3.4 System Design	12
3.5 Implementation	15
3.6 Platform	15
3.7 Objectives Of Internship	16
3.8 Learning Experiences	17
3.9 Learning Outcomes	18
4. RESUTS AND DISCUSSION	19-25
4.1 Project Results	19
4.2 Performance Evaluation	20-22
4.3 Advantages of the System	23
4.4 Limitations	24
4.5 Future Enhancements	25
CONCLUSION	26
REFERENCES	27

LIST OF FIGURES

FIGURE NO.	FIGURE CAPTION	PAGE NO.
3.4.1	Architecture Diagram of the Namma-Skill Android Application using Generative-AI	13
4.1	Terminal Output of Namma-Skill Application Execution and Testing	20
4.2	Traffic Flow Performance Metrics and Final	22

LIST OF TABLES

TABLE NO.	TABLE NAME	PAGE NO.
3.1	Work Schedule	6
3.3.1	Functional Requirements	9
3.3.2	Non-Functional Requirements	11
3.3.3	Software Requirements	12

CHAPTER 1

COMPANY PROFILE

1.1 Introduction

MindMatrix is an AI-powered educational and career development platform focused on transforming students into industry-ready professionals through modern technology-driven learning. The company bridges the gap between academic education and industry requirements by providing structured training, hands-on practical exposure, mentorship, and placement-oriented programs.

MindMatrix offers both online and offline learning environments that help students gain technical skills, problem-solving abilities, and real-world project experience. The platform is designed to support students, educational institutions, and organizations by integrating emerging technologies such as Artificial Intelligence, Cloud Computing, Android Development, Generative AI, and Industry 4.0 concepts into the learning process.

The company follows a systematic learning methodology called the **7P Framework**, which includes Presence, Perspective, Preparation, Practice, Profiling, Projection, and Progression. This framework helps students build professional identity, technical expertise, communication skills, and career readiness in a structured manner.

1.2 Mission

The mission of MindMatrix is to transform traditional education into a modern, technology-driven learning ecosystem that prepares students for real-world industry challenges. The company aims to empower students with AI-powered education, practical technical skills, innovation-oriented thinking, and career development opportunities that help them become industry-ready professionals.

MindMatrix focuses on bridging the gap between academic learning and industry expectations by providing hands-on training, live projects, mentorship, internships, and placement support. The company is committed to creating an engaging learning environment where students can develop problem-solving abilities, technical expertise, communication skills, and professional confidence.

1.2 Vision

The vision of MindMatrix is to become a leading AI-powered educational and career development platform that transforms the future of learning through innovation, technology, and industry integration. The company aims to create a skilled, future-ready workforce by bridging the gap between academic education and real-world industry requirement.

MindMatrix envisions building an advanced learning ecosystem where students can gain practical knowledge, technical expertise, creative thinking abilities, and professional confidence through AI-driven education and hands-on experience. The organization strives to make quality technical education accessible to students from all backgrounds, including rural and underserved communities.

1.3 Services

Mindmatrix provides a wide range of educational, technical, and career-oriented services designed to help students become industry-ready professionals through AI-powered and practical learning approaches.

- AI-Powered Learning Programs
- Online and Offline Training
- Learning Management System (LMS)
- Artificial Intelligence & Machine Learning
- Professional Skill Development
- Industry-Oriented Internships
- Placement & Talent Support

1.3.1 Executive Summary

Mindmatrix is an AI-powered educational and career development platform focused on transforming students into industry-ready professionals through technology-driven learning and practical skill development. The company provides both online and offline learning environments that integrate Artificial Intelligence, hands-on training, mentorship, internships, and placement support.

MindMatrix aims to bridge the gap between academic education and industry requirements by offering structured learning programs in domains such as Android App Development, Generative AI, Cloud Computing, Web Development, Salesforce, and

emerging technologies. The company follows a systematic learning methodology called the 7P Framework, which helps students build technical expertise, professional identity, innovation skills, and career readiness.

During the internship, students were exposed to modern technologies, AI tools, Android development concepts, cloud platforms, and real-world project implementation. The internship provided practical experience in Kotlin, Jetpack Compose, Firebase integration, MVVM architecture, API handling, and Generative AI integration through project-based learning.

As part of the internship project, an Android application named **Namma-Skill** was developed using Kotlin and Jetpack Compose. The application focuses on connecting rural youth with skill development and self-employment opportunities through an accessible and user-friendly mobile platform. The project integrates modern Android technologies and Generative AI concepts to provide intelligent recommendations, course exploration, and enrollment functionalities.

1.4 Conclusion

Mindmatrix plays an important role in transforming education through AI-powered learning, practical skill development, and industry-oriented training programs. The company successfully bridges the gap between academic education and real-world industry requirements by providing hands-on learning experiences, mentorship, internships, and placement support.

Through its innovative learning methodologies, modern technologies, and strong academia-industry collaborations, MindMatrix helps students develop technical expertise, professional confidence, and career readiness. The organization's focus on Artificial Intelligence, Generative AI, Android Development, Cloud Computing, and emerging technologies enables students to gain future-ready skills and practical exposure.

Overall, MindMatrix creates a supportive and technology-driven learning ecosystem that empowers students to become innovative, skilled, and industry-ready professionals capable of adapting to the rapidly evolving digital world.

CHAPTER 2

ABOUT COMPANY DEPARTMENT

2.1 Overview of the Department

The AI and Machine Learning department at MindMatrix consists of several functional components that work collaboratively to deliver a high-quality technical training and development experience. Each component holds specific responsibilities, ranging from the design of project assignments like the traffic flow optimization system to the management of curriculum delivery and practical guidance. While the training unit focuses on introducing modern frameworks and industry-relevant technologies, the Evaluation and Quality department ensures that all deliverables meet professional standards through a structured assessment of functionality and code quality. Additionally, the Support and Administration team manages essential logistical tasks, including intern onboarding, documentation, and the coordination of report submissions to ensure the smooth execution of the internship program.

2.2 Organizational Structure

The organizational structure of Mindmatrix follows a systematic hierarchy designed to support AI-powered education, technical training, internship management, and student career development. The structure ensures smooth coordination between academic learning, practical implementation, mentorship, and industry-oriented training activities.

- Management and Leadership Team
- Training and Learning Department
- Technical Trainers and Mentors
- Curriculum and Training Team
- Evaluation and Quality Team
- Project Assignment Team
- Administrative and Support Staff
- Student and Intern Cohort (Android Development, AI/ML, Web Development, Cloud, and Gen-AI domains)

All departments work collaboratively to maintain high standards of technical training, project quality, and skill development throughout the program. Technical trainers and mentors play a key role by guiding interns through the development lifecycle, helping them solve complex algorithmic challenges, and ensuring the effective implementation of deep learning models. This collaborative environment is supported by administrative staff who handle daily coordination and scheduling, allowing the intern cohort to focus on building functional and scalable AI systems.

CHAPTER 3

TASKS PERFORMED

3.1 Work Schedule

During the internship period at Mindmatrix, a structured learning schedule was followed to gain practical knowledge in Android application development, Generative AI, Firebase integration, and modern mobile technologies. The internship focused on both theoretical learning and hands-on implementation using Kotlin, Jetpack Compose, MVVM architecture, and AI-powered features.

Throughout the internship, the Namma-Skill application was developed as a real-world project to support skill development and self-employment opportunities. The following table summarizes the weekly tasks and activities performed during the internship period

Month	Week	Task Performed
Month 1	Week 1	Completed Google Cloud labs and explored internship learning platforms
Month 1	Week 2	Worked with Google AI Studio and developed AI-based prototype ideas.
Month 1	Week 3	Created Salesforce account and completed Trailhead Agentforce modules.
Month 1	Week 4	Configured service agents and earned badges through Trailhead activities.
Month 2	Week 5	Installed Android Studio, learned Kotlin basics, and developed simple applications.
Month 2	Week 6	Designed user interfaces using Jetpack Compose and built Android UI projects
Month 2	Week 7	Implemented advanced Kotlin concepts and developed interactive applications levels accurately.

Month 2	Week 8	Built multi-screen applications using MVVM architecture and View Model
Month 3	Week 9	Integrated APIs and implemented Room Database for data management
Month 3	Week 10	Implemented Work Manager features and performed testing and debugging activities.
Month 3	Week 11	Developed applications using XML UI, fragments, navigation, and integrated Jetpack Compose into Android projects.
Month 3	Week 12	Completed Agent force Super badge, explored prompt engineering, and worked on AI agent testing and data integration

Table 3.1 Work Schedule

Table 3.1 summarizes the complete work schedule followed during the internship period. The work was divided into three structured months focusing on data acquisition, deep learning model development, and system simulation. The initial phase focused on defining traffic optimization objectives, collecting video data from open datasets, and performing image preprocessing tasks such as resizing and noise reduction. The second phase involved implementing vehicle detection using deep learning models, classifying traffic density, and training predictive models to assess congestion levels. The final phase included developing dynamic signal timing logic, simulating traffic control through software, and evaluating the system using specific flow metrics. This structured approach ensured systematic development, a practical understanding of Artificial Intelligence and Machine Learning, and the successful completion of the project.

3.2 Introduction

The project titled **“Namma-Skill – Self Employment and Skill Development Platform”** is an Android-based mobile application developed to support rural youth and job seekers by providing easy access to skill development and self-employment opportunities. The application aims to bridge the gap between unemployed individuals and available vocational training programs through a simple, accessible, and user-friendly digital platform.

Many individuals, especially in rural areas, face difficulties in finding information about training centers, skill-based courses, and career development opportunities due to limited awareness and the absence of centralized platforms. This project addresses that challenge by creating an Android application where users can explore, search, and enroll in training programs efficiently.

The application is developed using Kotlin and Android Studio with Jetpack Compose for modern UI design. Firebase or Room Database is used for data management, while Generative AI technologies through Google AI Studio are integrated to provide intelligent recommendations and improve user interaction. The project includes features such as course browsing, category-based filtering, training center details, notifications, and enrollment functionality.

The project not only focuses on improving digital accessibility and employment opportunities but also demonstrates the practical implementation of Android application development, UI/UX design, database integration, and Generative AI concepts in a real-world solution.

3.3 System Requirements

The Namma-Skill application requires a proper combination of hardware resources, software tools, and modern development technologies to ensure smooth performance and efficient functionality. The system is designed to provide a user-friendly Android platform that helps users explore skill development programs, training opportunities, and self-employment resources through digital technology.

The application includes several functional requirements such as course browsing, searching and filtering programs, viewing training center details, enrollment management, and notification services. In addition to these functionalities, the system also focuses on non-functional requirements including fast performance, reliability, user-friendly interface design, secure data handling, and compatibility with Android devices.

The project is developed using Android Studio and Kotlin with Jetpack Compose for modern UI development. Firebase or Room Database is used for efficient data storage and retrieval, while Google AI Studio and Generative AI technologies are integrated to provide

intelligent recommendations and enhanced user interaction. The application is designed to support scalable development so that future enhancements such as real-time location tracking, AI-based personalization, and advanced analytics can be integrated easily for modern UI development.

Firebase or Room Database is used for efficient data storage and retrieval, while Google AI Studio and Generative AI technologies are integrated to provide intelligent recommendations and enhanced user interaction. The application is designed to support scalable development so that future enhancements such as real-time location tracking, AI-based personalization, and advanced analytics can be integrated easily.

To ensure smooth execution and development, the system requires Android devices or computers with sufficient processing capability, memory, storage, and internet connectivity. These system requirements help maintain stable application performance, efficient testing, and reliable user experience throughout the project lifecycle

3.3.1 Functional Requirements

Table 3.2 Functional Requirements

Functionality	Description
User Registration and Login	The system shall allow users to create accounts and securely log in to the application.
Course Browsing	The system shall provide users with the ability to browse available skill development and self-employment courses.
Search and Filtering	The application shall allow users to search and filter courses based on category, duration, and training type.
Training Center Information	The system shall display details of nearby training centers and available programs.
Course Enrollment	The system shall enable users to apply and enroll in training programs directly through the application.
AI-Based Recommendations	The application shall provide intelligent course recommendations using Generative AI technologies.
Notifications and Updates	The system shall send notifications regarding new training batches, course updates, and enrollment confirmations.
User Profile Management	The application shall allow users to manage personal profiles, course history, and preferences.

Data Storage and Retrieval	The system shall efficiently store and retrieve user data, course details, and enrollment information using Firebase or Room Database.
Secure Data Handling	The system shall ensure secure handling and management of user information and application data.
Responsive User Interface	The application shall provide a user- friendly and responsive Android interface for smooth user interaction.

Table 3.2 presents the core functional requirements of the Namma-Skill application. These requirements define the essential operations needed to support skill development and self-employment opportunities through a structured and user-friendly Android platform. The system focuses on functionalities such as user registration, course browsing, search and filtering, enrollment management, notifications, and AI-based recommendations to ensure smooth user interaction and efficient application performance.

3.3.2 Non-Functional Requirements

Table 3.3 Non-Functional Requirements

Requirement	Description
Real-Time Performance	The system shall process video frames and update signal timing logic within a minimal delay to maintain current flow.
Detection Accuracy	The deep learning model shall maintain high precision in vehicle counting and classification to prevent incorrect signal adjustments.
Reliability	The system shall ensure consistent operation and data handling during continuous video processing without system crashes.
Scalability	The modular architecture shall allow the addition of more camera feeds or integration with broader smart city networks in the future.
Usability	The simulation and reporting interface shall provide clear visual data, making it accessible for traffic administrators to monitor road conditions.

Table 3.2 presents the functional requirements of the Namma-Skill application. These requirements define the core operations needed to support skill development and self-employment opportunities through a user-friendly Android platform.

The system includes features such as user registration, course browsing, search and filtering, course enrollment, notifications, and AI-based recommendations. These functionalities ensure smooth performance, efficient data management, and improved user experience throughout the application

3.3.3 Software Requirement

Table 3.4 Software Requirements

Component	Specification
Development IDE	Android Studio / Visual Studio Code
Programming Language	Kotlin
UI Framework	Jetpack Compose
Architecture Pattern	MVVM Architecture
Database	Firebase / Room Database
AI Integration Tool	Google AI Studio
Build System	Gradle

Table 3.4 shows the software tools, frameworks, and technologies used for developing and running the Namma-Skill application. The selected technologies ensure efficient Android application development, smooth user interaction, secure data management, and reliable Generative AI integration for providing skill development and self-employment support through a modern mobile platform.

3.4 System Design

The Namma-Skill application is designed as an intelligent and user-friendly Android platform that supports skill development and self-employment opportunities for users. The system follows a modular Android application architecture integrated with Generative AI technologies to provide efficient course management, user interaction, and personalized recommendations.

The design separates different functionalities such as user authentication, course browsing, search and filtering, enrollment management, notifications, and AI-based recommendation modules to ensure smooth application performance and maintainability. The application is developed using Kotlin, Jetpack Compose, and MVVM architecture to provide a responsive, scalable, and modern mobile user experience.

The system design focuses on efficient data management, secure user interaction, and seamless navigation throughout the application. Firebase or Room Database is used for storing and retrieving user and course information, while Generative AI integration improves recommendation accuracy and user engagement. The interface is designed to be

simple, interactive, and accessible, enabling users to easily explore training opportunities, manage enrollments, and receive real-time updates through an intuitive mobile environment.

3.4.1 System Architecture

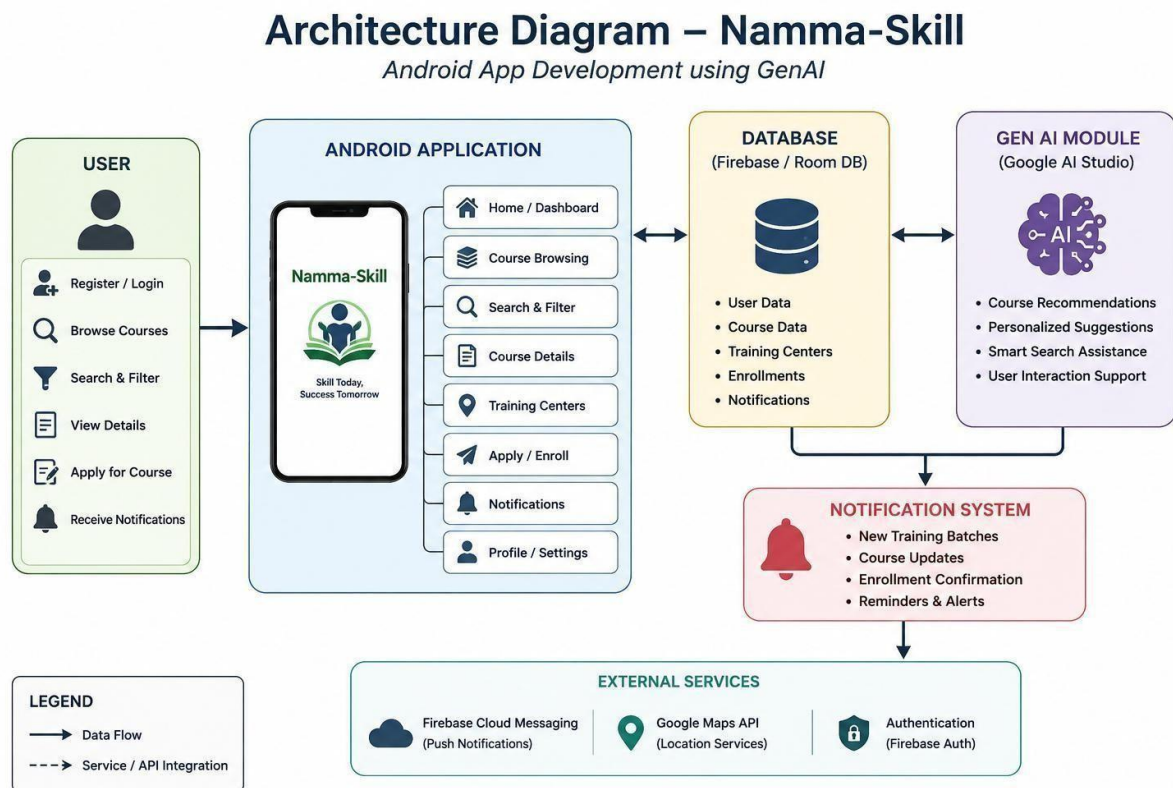


Figure 3.4.1: Architecture Diagram of the Namma-Skill Android Application using Generative AI

The architecture of the proposed **Namma-Skill Application** is illustrated in Figure 3.4.1. The system is designed using a modular Android application architecture integrated with Generative AI technologies to provide an efficient, scalable, and user-friendly platform for skill development and self-employment support.

The system begins with the **User Layer**, where users interact with the application through functionalities such as registration/login, course browsing, search and filtering, viewing course details, applying for courses, and receiving notifications. This layer acts as the primary interface between users and the application.

The core functionality is handled by the **Android Application Layer**, developed using Kotlin, Jetpack Compose, and MVVM architecture. This layer manages modules such as the dashboard, course browsing, search and filtering, training center details,

enrollment management, notifications, and user profile settings. The modular design ensures smooth navigation, responsive UI, and efficient application performance.

The application communicates with the **Database Layer**, implemented using Firebase or Room Database. This layer stores and manages user information, course details, training center data, enrollments, and notification records. Efficient data storage and retrieval mechanisms help maintain reliable system performance and secure user data handling.

The system also integrates a **Generative AI Module** using Google AI Studio. This layer provides intelligent functionalities such as personalized course recommendations, smart search assistance, and enhanced user interaction support. AI integration improves user engagement and helps users discover suitable training opportunities based on their interests and activities.

A separate **Notification System** is included to deliver updates related to new training batches, course changes, enrollment confirmations, reminders, and alerts. This improves communication and keeps users informed about available opportunities.

Finally, the architecture connects with **External Services** such as Firebase Cloud Messaging for push notifications, Google Maps API for location-based services, and Firebase Authentication for secure user login and account management. Together, these integrated layers ensure efficient application functionality, intelligent recommendations, secure data handling, and a seamless user experience throughout the Namma-Skill platform.

3.4.2 Working Flow of System

1. The user registers or logs into the Namma-Skill application using secure authentication.
2. The system displays the home dashboard with available skill development and self-employment programs.
3. The user browses courses and training opportunities through the Android application interface.
4. The system allows users to search and filter courses based on category, duration, and preferences.
5. The application retrieves course and training center data from Firebase or Room Database.
6. The Generative AI module analyzes user interests and provides personalized course recommendations.

7. The user views detailed information about courses and nearby training centers.
8. The system allows users to apply and enroll in selected training programs directly through the application.
9. The notification system sends updates regarding enrollments, new batches, reminders, and course announcements.
10. The application securely stores user activities, enrollment details, and preferences for future access and improved recommendations.

3.5 Implementation

The Namma-Skill application was implemented using Kotlin and Android Studio with Jetpack Compose for modern Android UI development. The system follows MVVM architecture for efficient management of user interface, business logic, and data handling.

Firebase or Room Database was used for storing user and course information, while Google AI Studio was integrated to provide AI-based recommendations and smart assistance. Features such as user authentication, course browsing, search and filtering, enrollment, notifications, and profile management were successfully implemented.

The application was tested using Android Emulator and Android devices to ensure smooth performance, responsive design, and reliable user experience.

3.6 Platform

The Namma-Skill application was developed using modern Android development technologies, Generative AI tools, databases, and frameworks to provide a user-friendly and intelligent mobile platform for skill development and self-employment support. These platforms were used for designing responsive user interfaces, implementing AI-powered recommendations, managing user data, and ensuring smooth application performance.

3.6.1 Programming Languages

The following programming languages were used during the development of the Vehicle Rental Management System:

- **Kotlin** – Used as the primary programming language for Android application development, UI handling, business logic, and application functionalities.
- **SQL**: Used for managing and querying traffic flow metrics and historical congestion data.
- **YAML**: Used for configuring model hyperparameters and dataset directory structures for the detection framework.

3.6.2 Libraries and Tools

These libraries and tools collectively support efficient Android application development, AI integration, secure data management, and smooth user interaction. The following tools and technologies were used for project implementation:

- **Android Studio** – Used as the primary development environment for building, testing, and debugging the Android application.
- **Jetpack Compose** – Used for developing modern, responsive, and interactive Android user interfaces.
- **Firebase** – Used for authentication, cloud messaging, notifications, and data management functionalities.
- **Room Database** – Used for efficient local data storage and retrieval within the application.
- **Google AI Studio** – Used for integrating Generative AI features such as personalized recommendations and smart assistance.
- **MVVM Architecture** – Used to separate UI, business logic, and data handling for scalable and maintainable application development.
- **Gradle** – Used for dependency management and application build configuration.
- **Android Emulator** – Used for testing and validating application performance on virtual Android devices.
- **Google Maps API** – Used for location-based training center and navigation support.

The use of structured Android development practices ensures smooth application performance, secure data handling, efficient navigation, and reliable AI integration. The modular architecture also simplifies maintenance, testing, debugging, and future feature enhancements such as advanced AI recommendations, multilingual support, and additional employment services.

3.7 Objectives Of Internship

- To gain hands-on experience in Android application development using Kotlin and Jetpack Compose.
- To understand the implementation of MVVM architecture and modern Android development practices.
- To develop practical knowledge of Generative AI integration using Google AI Studio.
- To learn database management using Firebase and Room Database for mobile

applications.

- To design and develop responsive and user-friendly Android application interfaces.
- To implement features such as course browsing, search and filtering, enrollment, and notifications in a real-world application.
- To improve problem-solving, debugging, testing, and application optimization skills.
- To gain experience in project planning, documentation, and software development lifecycle practices.
- To strengthen technical skills in Android Studio, Gradle, APIs, and modern mobile development tools.
- To enhance teamwork, communication, and industry-oriented development experience through internship activities.

3.8 Learning Experiences

- **Android Development Skills:** Gained practical experience in Android application development using Kotlin and Android Studio.
- **Jetpack Compose UI Design:** Learned to create modern, responsive, and interactive mobile user interfaces using Jetpack Compose.
- **MVVM Architecture:** Understood the implementation of MVVM architecture for better project structure, scalability, and maintainability.
- **Generative AI Integration:** Explored AI-powered features using Google AI Studio for intelligent recommendations and user interaction.
- **Database Management:** Learned data storage and retrieval using Firebase and Room Database for efficient application performance.
- **Navigation and State Management:** Developed skills in implementing navigation systems, fragments, and state handling within Android applications.
- **API and Feature Integration:** Gained experience integrating APIs, notifications, and location-based services into mobile applications.
- **Testing and Debugging:** Improved debugging, testing, and performance optimization skills using Android Emulator and development tools.
- **Project Development Experience:** Learned the complete software development lifecycle including planning, implementation, testing, and documentation.
- **Version Control and Collaboration:** Strengthened knowledge of Git and GitHub for code management, project collaboration, and version tracking.

3.9 Learning Outcomes

- **Android Development Proficiency:** Developed practical knowledge in building Android applications using Kotlin and Android Studio.
- **Modern UI Development Skills:** Gained experience in designing responsive and interactive user interfaces using Jetpack Compose.
- **MVVM Architecture Understanding:** Learned to implement MVVM architecture for structured, scalable, and maintainable application development.
- **Generative AI Knowledge:** Acquired understanding of AI integration using Google AI Studio for intelligent recommendations and user assistance.
- **Database Management Skills:** Learned efficient data storage and retrieval using Firebase and Room Database.
- **API and Feature Integration:** Gained experience in integrating APIs, notifications, and location-based functionalities into Android applications.
- **Testing and Debugging Skills:** Improved the ability to identify, debug, and resolve application errors and performance issues.
- **Software Development Lifecycle Experience:** Understood the complete process of planning, designing, implementing, testing, and documenting software projects.
- **Project Management and Collaboration:** Enhanced teamwork, communication, and version control skills using Git and GitHub.
- **Industry-Oriented Technical Exposure:** Gained real-world exposure to modern Android development tools, AI technologies, and professional development practices.
- **Professional Workflow Management:** Improved proficiency in using industry-standard tools like GitHub and Visual Studio Code for version control and development.

CHAPTER 4

RESULTS

The Namma-Skill application was successfully developed and tested as an Android-based platform designed to support skill development and self-employment opportunities through a user-friendly and AI-powered mobile environment. The final application demonstrated effective course browsing, search and filtering, enrollment management, notification services, and personalized AI-based recommendations.

The system successfully provides users with easy access to training programs, skill development resources, and self-employment opportunities through an interactive and responsive Android application. The integration of Kotlin, Jetpack Compose, Firebase, MVVM architecture, and Generative AI technologies ensured smooth application performance, efficient data management, secure user interaction, and improved user experience.

The project successfully achieved its objective of creating a modern digital platform that bridges the gap between users and available training opportunities while demonstrating practical implementation of Android development and AI-powered application features.

4.1 Project Results

The implemented Namma-Skill application successfully provides a functional and user-friendly Android platform that allows users to explore skill development and self-employment opportunities efficiently. The application enables users to browse available courses, search and filter training programs, view training center details, and enroll in courses through an interactive mobile interface.

The final output demonstrated successful implementation of features such as user authentication, course management, AI-based recommendations, notifications, and profile handling. The integration of Kotlin, Jetpack Compose, Firebase, and Generative AI technologies ensured smooth application performance, responsive navigation, secure data management, and an improved user experience.

The project successfully achieved its objective of creating a digital platform that

bridges the gap between users and available training opportunities while providing practical implementation of Android application development and AI-powered functionalities.

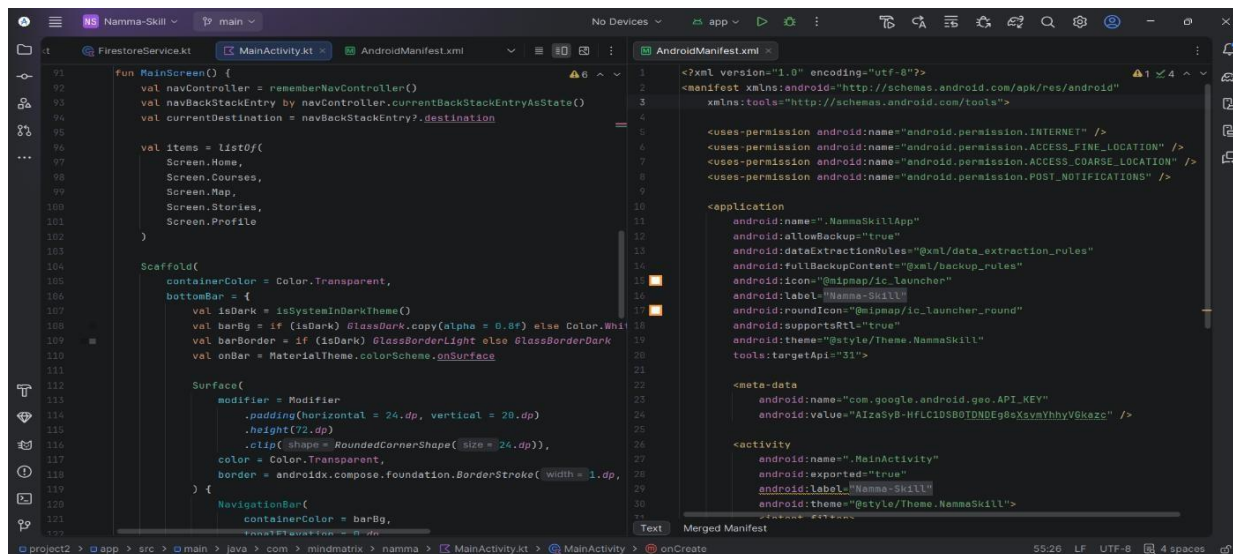


Figure 4.1: Main Activity and Manifest file of Namma-Skill Application

4.2 Performance Evaluation

The performance of the Namma-Skill application was evaluated based on application responsiveness, navigation efficiency, data handling, and user interaction. The testing process verified that the application functions smoothly across different Android environments while maintaining secure and reliable data management.

4.2.1 Application Performance Metrics

UI Responsiveness: The application provided smooth navigation between screens such as login, dashboard, course listing, search, enrollment, and profile management with minimal loading delays.

Database Performance: Firebase or Room Database successfully handled user data storage, retrieval, and enrollment information efficiently without data loss.

AI Recommendation Functionality: The Generative AI module successfully provided intelligent recommendations and improved user interaction within the application.

Navigation Efficiency: The MVVM architecture and Jetpack Compose navigation system ensured organized screen transitions and stable application performance.

4.2.2 System Results

Successful Android Application Execution

The application was successfully built and executed using Android Studio and Android Emulator.

User Authentication:

The system successfully supported user registration and login functionalities.

Course Management:

Users were able to browse, search, filter, and enroll in training programs effectively.

Notification System:

The application successfully displayed notifications and updates related to courses and enrollments.

Stable System Performance:

The application maintained stable performance and responsive user interaction during testing and execution.

Results Tested

The testing phase of the Namma-Skill application was conducted to verify the functionality, responsiveness, and reliability of the Android application and its AI-powered features. The following results were observed during system execution:

- **Successful Application Build and Execution:** The application was successfully built and executed using Android Studio and Android Emulator without major runtime issues.
- **User Authentication Functionality:** The system successfully handled user registration and login processes with secure authentication support.
- **Smooth Navigation and UI Performance:** The Jetpack Compose interface provided responsive navigation between screens such as dashboard, course listing, search, enrollment, notifications, and profile management.
- **Efficient Course Browsing and Filtering:** Users were able to browse, search, and filter training programs efficiently based on categories and preferences.
- **Database Integration Success:** Firebase or Room Database successfully stored and

retrieved user data, course information, and enrollment records without data loss.

- **AI-Based Recommendation Support:** The Generative AI module successfully provided intelligent recommendations and enhanced user interaction within the application.
- **Notification System Functionality:** The application successfully delivered notifications related to new courses, enrollments, and updates.
- **System Stability and Responsiveness:** The application maintained stable performance, quick response time, and smooth user interaction during testing and execution.

These results demonstrate that the Namma-Skill application successfully provides an efficient, user-friendly, and AI-powered platform for skill development and self-employment support through modern Android technologies.

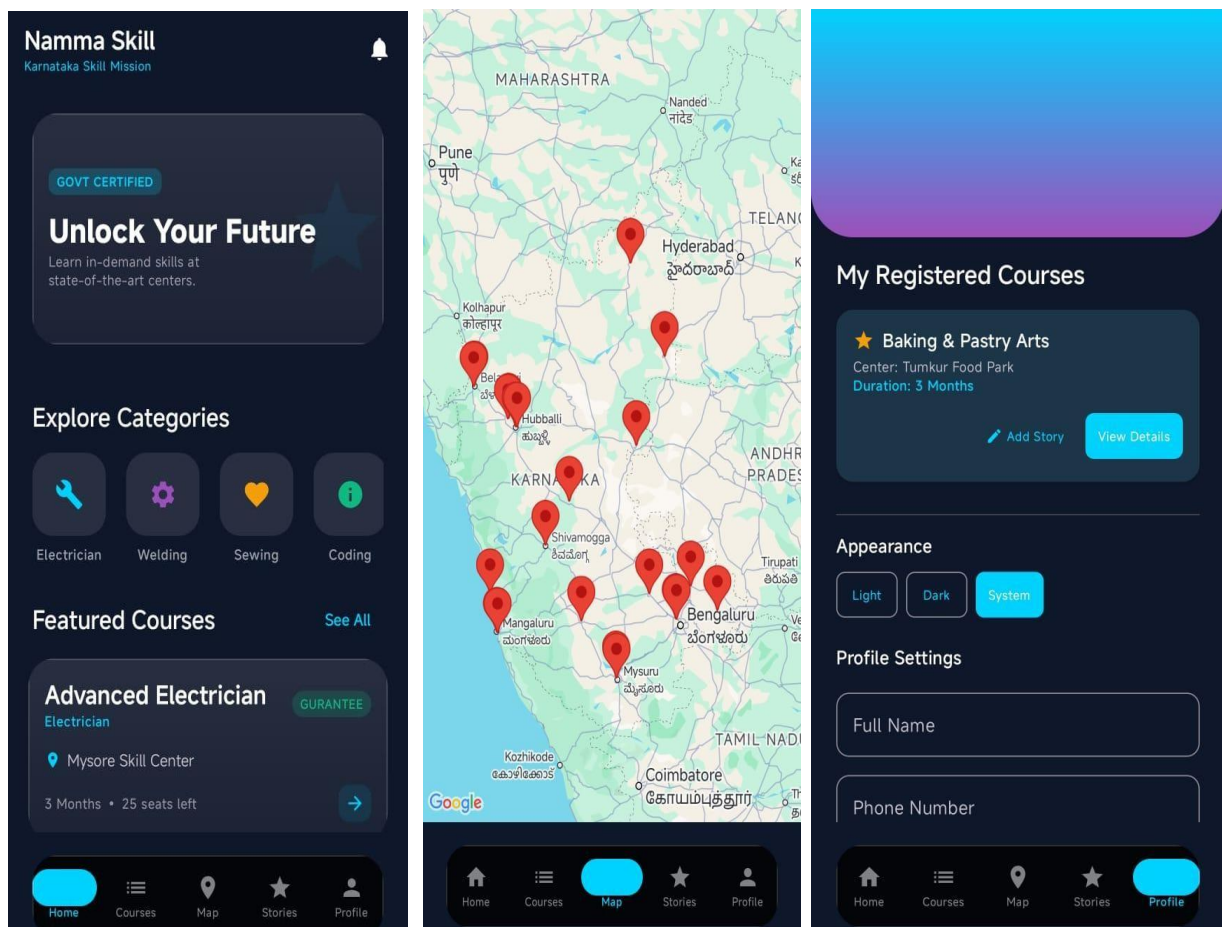


Figure 4.2: Namma Skill user dashboard

This figure illustrates the successful execution and user interface performance of the Namma-Skill Android application. The displayed dashboard represents the main home screen of the application, where users can explore skill development opportunities, search

courses, and access AI-powered recommendations through a modern and user-friendly interface.

The upper section of the application displays personalized user information, including location details and profile access, along with an integrated search functionality for quickly finding skill-based courses and training programs. The categorized sections such as Technical, Design, and Agro provide simplified navigation for users to explore different domains of learning and self-employment opportunities.

The middle section highlights the AI-powered recommendation system, which suggests relevant courses such as Mobile Repairing and Tailoring based on user interests and application functionality. Course cards display important details including course category, training center information, distance, and pricing, ensuring an organized and interactive user experience. The lower section demonstrates the integrated AI Career Path module and bottom navigation system, which provide easy access to core application features such as Home, Courses, Notifications, and Profile management. The application maintained smooth responsiveness, efficient navigation, and stable performance during testing on Android Emulator.

Overall, this figure validates the successful implementation of the Namma-Skill application and demonstrates the effectiveness of Kotlin, Jetpack Compose, Firebase integration, and Generative AI features in creating a responsive and scalable Android-based skill development platform.

4.3 Advantages of the System

- **User-Friendly Interface:** The application provides a simple, modern, and easy-to-use interface for users to explore skill development opportunities efficiently.
- **AI-Based Recommendations:** Integration of Generative AI helps provide intelligent course suggestions and personalized user interaction.
- **Easy Course Browsing and Filtering:** Users can quickly search, filter, and access training programs based on categories and preferences.
- **Secure User Authentication:** Firebase authentication ensures secure login, registration, and user data management.
- **Real-Time Notifications:** The system provides timely updates regarding course enrollments, new training batches, and notifications.
- **Efficient Database Management:** Firebase and Room Database support reliable storage and retrieval of user and course information.

- **Responsive Android Application:** Jetpack Compose and MVVM architecture ensure smooth navigation and responsive application performance.
- **Scalable System Architecture:** The modular application design allows easy maintenance, future feature integration, and scalability.
- **Supports Skill Development and Self-Employment:** The platform helps users access learning opportunities and improve career and self-employment prospects.
- **Cross-Functional Technology Integration:** The project demonstrates successful integration of Android development, database management, AI technologies, and modern UI design practices.

4.4 Limitations

- **Internet Dependency:** Some features such as Firebase synchronization, AI recommendations, and notifications require a stable internet connection.
- **Limited Offline Functionality:** Certain application functionalities may not be fully accessible in offline mode.
- **Device Performance Variations:** Application performance may vary on low-end Android devices with limited processing power or memory.
- **AI Recommendation Accuracy:** The quality of AI-based recommendations depends on available user data and interaction history.
- **Database Dependency:** Proper functionality depends on stable Firebase or database configuration and connectivity.
- **Location Service Dependency:** Features related to nearby training centers rely on GPS and location permissions being enabled.
- **Scalability Challenges:** Handling a large number of simultaneous users may require advanced backend optimization and server scaling.
- **Security and Privacy Concerns:** User data management requires continuous security updates and proper authentication handling to maintain data privacy.

4.5 Future Enhancements

- **Advanced AI Recommendations:** Implement more intelligent AI models for personalized course and career recommendations based on user interests and learning history.
- **Multilingual Support:** Add support for multiple regional languages to improve accessibility for users from different backgrounds.
- **Online Certification System:** Provide digital certificates for users after successful completion of training programs and courses.
- **Video Learning Integration:** Include video-based tutorials, live classes, and recorded learning sessions within the application.
- **Real-Time Chat Support:** Integrate chat and mentorship features for direct communication between trainers and users.
- **Job and Internship Recommendations:** Add features to suggest jobs, internships, and self-employment opportunities based on user skills and completed courses.
- **Offline Learning Support:** Enable offline access to selected courses and training materials for users with limited internet connectivity.
- **AI Chatbot Assistance:** Introduce an AI-powered chatbot to guide users in selecting courses and resolving application-related queries.
- **Government Scheme Integration:** Integrate government skill development and employment programs to improve accessibility to official training opportunities.
- **Advanced Analytics Dashboard:** Develop analytics and progress tracking features to monitor user learning activities, enrollments, and achievements.

4.6 Conclusion

The Namma-Skill application successfully addresses the challenge of providing accessible skill development and self-employment opportunities through a modern Android-based platform. By integrating Kotlin, Jetpack Compose, Firebase, MVVM architecture, and Generative AI technologies, the application demonstrates an efficient and user-friendly solution for course exploration, enrollment management, notifications, and personalized recommendations.

The project successfully implemented important features such as user authentication, course browsing, category-based filtering, AI-powered recommendations, and responsive UI navigation. The application maintained stable performance during testing and provided smooth interaction across different modules within the Android environment.

Overall, the project serves as a scalable and practical solution for promoting digital learning and self-employment opportunities. The integration of modern Android technologies and AI-powered functionalities highlights the real-world applicability of the system while providing a strong foundation for future enhancements in mobile learning and career development platforms.

REFERENCES

1. Google. *Android Developers Documentation*. Available at: <https://developer.android.com>
2. Kotlin. *Official Kotlin Documentation*. Available at: <https://kotlinlang.org>
3. Firebase. *Firebase Documentation*. Available at: <https://firebase.google.com/docs>
4. Jetpack Compose. *Android UI Toolkit Documentation*. Available at: <https://developer.android.com/jetpack/compose>
5. Google AI Studio. *Generative AI Platform*. Available at: <https://aistudio.google.com>
6. Room Database. *Android Room Persistence Library*. Available at: <https://developer.android.com/training/data-storage/room>
7. Android Studio. *Official Android IDE*. Available at: <https://developer.android.com/studio>
8. GitHub. *Version Control Platform*. Available at: <https://github.com>
9. Department of AIML, Internship Report Documentation, 2025–26.